Introduction to Database Management
Unit-1

Data: Data is defined facts/figures.
Fact: Something that really exist.
Example: 502
Data is abstract in nature (Raw facts).
Raw fact: Raw fact means that the facts have not yet been proceed to get their exact meaning.
Why we need Data: To derive some information from it.

Information: Information is organized or classified data, which has some meaningful values for the receiver.
Example: 502 is a room number.

Why we process data: For future analysis.

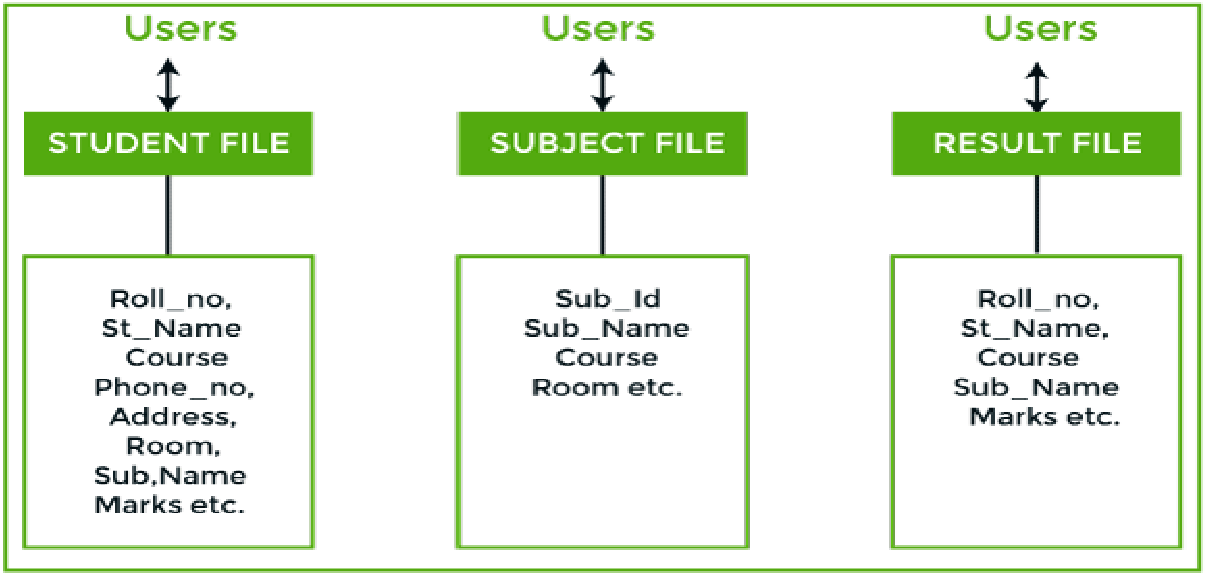
Record: Collection of **inter-related data.**
Inter-related data: That the data is connected and related to each other with respect to the given **attributes.**

Attributes: It is a quality/ characteristics of person, place or real things.
Example: Employee id is related to the name, salary, department number, etc.

File Oriented approach:

- File based systems were an early attempt to computerize the manual system.
- It is also called a traditional based approach in which a decentralized approach was taken where each department stored and controlled its own data with the help of a data processing specialist.

The main role of a data processing specialist was to create the necessary computer file structures, and also manage the data within structures and design some application programs that create reports based on file data.
Example: Consider an example of a student's file system. The student file will contain information regarding the student (i.e. roll no, student name, course etc.). Similarly, we have a subject file that contains information about subject and result file which contains the information regarding the result.



Some fields are duplicated in more than one file, which leads to data redundancy. So to overcome this problem, we need to create a centralized system, i.e. DBMS approach.

Disadvantages of file oriented approach:

1) Data redundancy and inconsistency:

File processing system leads to the usage of many copies of same data. If we need to change any of the data, then we need to change the data at all copies. If not, this will lead to inconsistency.

Example: Assume a file for storing addresses of students. If we make three copies of the address file and store them in three different computers, we say that the data is redundant. If suppose one want to change the address of any students, then the change should be made at all the three computers failing which leads to inconsistent data.

2) Difficulty in accessing data :

In a file processing system, to access data differently we need to have different programs.

Example: If you want to access student names from a file, we need a program that does the job. If you want to view only address of all students from a specific city, then we need different program that does the required job. This list goes endless. Hence, it is difficult to access data.

3) Data isolation :

Because data are scattered in various file and files may be in different formats, with new application programs to retrieve the appropriate data is difficult.

Example: At one location the student data may be stored in **.txt** format. In other location, the same file may be stored in **.doc** format.

4) Integrity Problems:

Developers enforce data validation in the system by adding appropriate code in the various application program. However when new constraints are added, it is difficult to change the programs to enforce them.

5) Atomicity:

It is difficult to ensure atomicity in a file processing system when transaction failure occurs due to power failure, networking problems etc.

(atomicity: either all operations of the transaction are reflected properly in the database or no changes are reflected.)

6) Concurrent access:

In file processing system it is not possible to access same file for multiple transactions at same time .

Example: If only one ticket is there and two customers are trying to book the ticket simultaneously, ticket should be allotted to any one customer.

7) Security problems:

There is no security provided in file processing system to secure the data from unauthorized user access.

Database: The database is a collection of inter-related data which is used to retrieve, insert and delete the data efficiently. It is also used to organize the data in the form of a table, schema, views, and reports, etc.

The **main purpose** of the database is to operate a large amount of information by storing, retrieving, and managing data.

Example: The college Database organizes the data about the admin, staff, students and faculty etc.

The main features of data in a database are:

1. It must be well organized
2. It is related
3. It is accessible in a logical order without any difficulty
4. It is stored only once

Why Database:

In order to overcome the limitation of a file system, a new approach was required. Hence a database approach emerged. A database is a persistent collection of logically related data. The initial attempts were to provide a centralized collection of data. A database has a self describing nature. It contains not only the data sharing but integration of data of an organization in a single database.

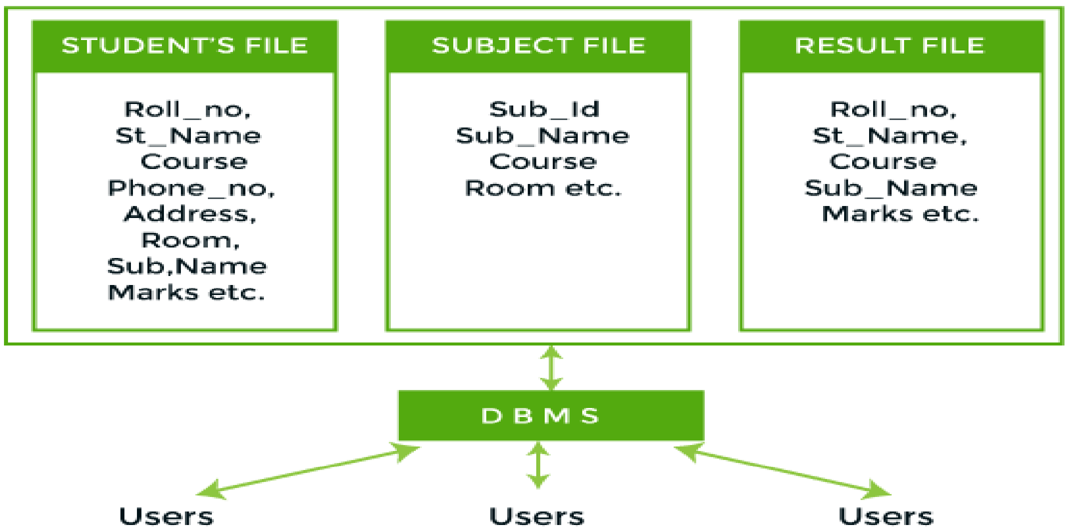
A small database can be handled manually but for a large database and having multiple users it is difficult to maintain it, In that case a computerized database is useful. The advantages of database system over traditional, paper based methods of record keeping are:

- **compactness:** No need for large amount of paper files
- **speed:** The machine can retrieve and modify the data more faster way then human being
- **Less drudgery:** Much of the maintenance of files by hand is eliminated

- **Accuracy:** Accurate, up-to-date information is fetched as per requirement of the user at any time.

  Example:

  In figure, duplication of data is reduced due to centralization of data.



Database System : It provides a convenient way to store, retrieve as well as to manipulate the data.

Database Management System (DBMS):

- DBMS is a collection of program that enables user to create and maintain a database.
- Database management system is a software which is used to manage the database.

Example: MySQL, Oracle are a very popular commercial database which is used in different applications.

- DBMS provides an interface to perform various operations like database creation, storing data in it, updating data, creating a table in the database and a lot more.
- It provides protection and security to the database. In the case of multiple users, it also maintains data consistency.

Characteristics of DBMS

- Self describing
- Multiple view support
- Data sharing
- Integrity
- Multiple user transaction processing
- Data independence
- Minimal redundancy
- Privacy and security

Function of DBMS:

1. Defining database schema: It must give facility for defining the database structure also specifies access rights to authorized users.

2. Manipulation of the database: The dbms must have functions like insertion of record into database updation of data, deletion of data, retrieval of data

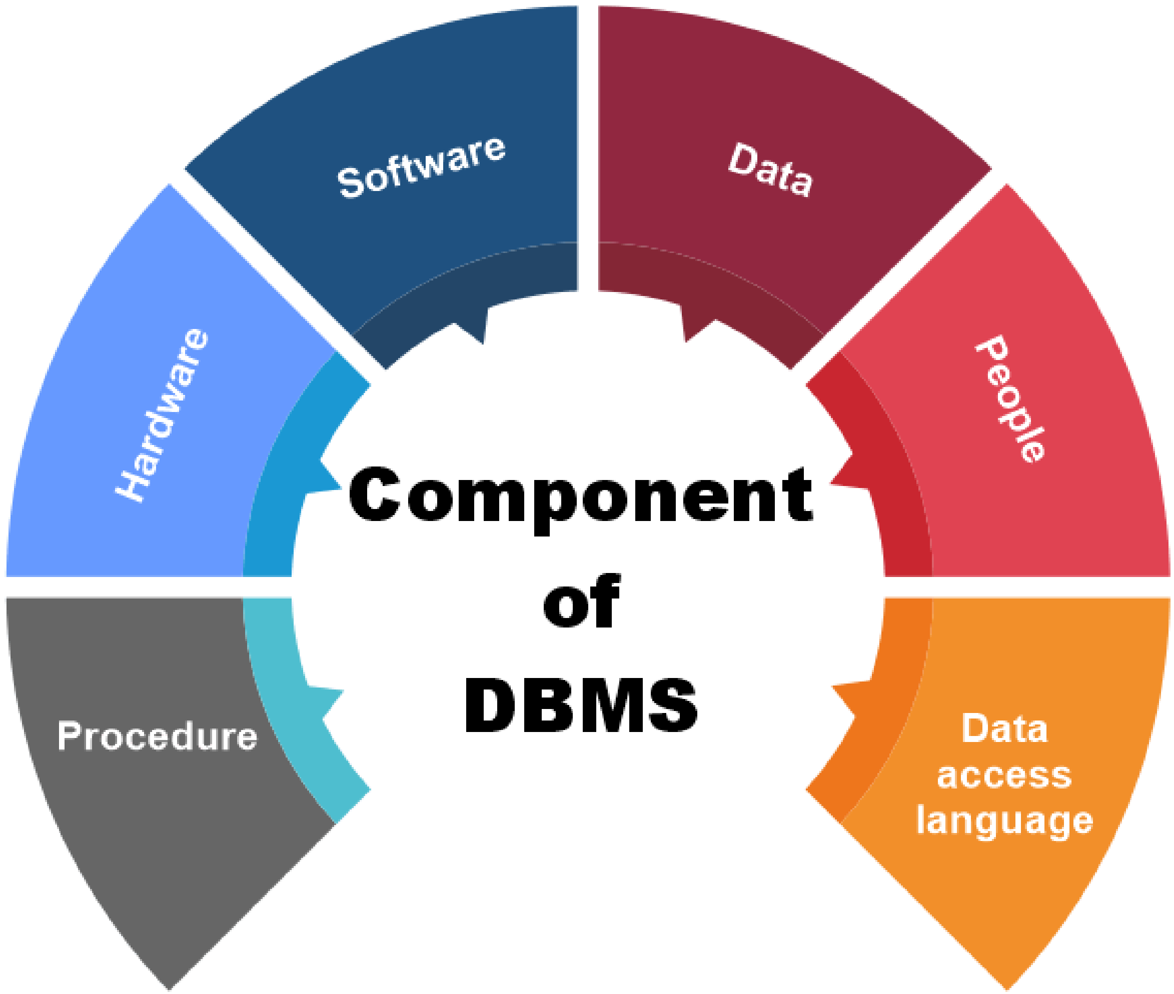
3. Sharing of database: The DBMS must share data items for multiple users by maintaining consistency of data.

4. Protection of database: It must protect the database against unauthorized users.

5. Database recovery: If for any reason the system fails DBMS must facilitate data base recovery.

Components of DBMS

There are many components available in the DBMS. These components consist of people, the technique of Handel the database, data, hardware, software, etc.



1. Hardware

- Hardware means the physical part of the DBMS. Here the hardware includes output devices like a printer, monitor, etc., and storage devices like a hard disk.
- In DBMS, information hardware is the most important visible part. The equipment which is used for the visibility of the data is the printer, computer, scanner, etc. This equipment is used

to capture the data and present the output to the user.

- With the help of hardware, the DBMS can access and update the database.

2. Software

- Software is defined as the collection of programs that are used to instruct the computer about its work. The software consists of a set of procedures, programs, and routines associated with the computer system's operation and performance.
- The software includes so many software like network software and operating software. The database software is used to access the database, and the database application performs the task.
- This software has the ability to understand the database accessing language and then convert these languages to real database commands and then execute the database.
- Some examples of DBMS software include MySQL, Oracle, SQL Server, dBase,Microsoft Access, etc.

3.Data

- The term data means the collection of any raw fact stored in the database. Here the data are any type of raw material from which meaningful information is generated.
- The database can store any form of data, such as structural data, non-structural data, and logical data.
- With the help of the database, we can create and construct the DBMS. After the creation of the database, we can create, access, and update that database.
- When the user stores the data in a database, such as the size of the data, the name of the data, and some data related to the user, these data are called metadata.

4. Procedures

- The procedure is a type of general instruction or guidelines for the use of DBMS. This instruction includes how to set up the database, how to install the database, how to log in and log out of the database, how to manage the database, how to take a backup of the database, and how to generate the report of the database.

- With the help of procedure, we can validate the data, control the access and reduce the traffic between the server and the clients.
- The main purpose of the procedure is to guide the user during the management and operation of the database.

5. Database Access Language

- Database Access Language is a simple language that allows users to write commands to perform the desired operations on the data that is stored in the database.
- It is used to write commands to access, insert, update and delete data stored in a database.
- Examples of database languages are SQL (structured query language), My Access, Oracle, etc. A database language is comprised of two languages.

6. People/Users

- The people who control and manage the databases and perform different types of operations on the database in the DBMS.
- The people include database administrator, software developer, and End-user.
- Database administrator-database administrator is the one who manages the complete database management system. DBA takes care of the security of the DBMS, its availability, managing the license keys, managing user accounts and access, etc.
- Software developer- This user group is involved in developing and designing the parts of DBMS. They can handle massive quantities of data, modify and edit databases, design and develop new databases, and troubleshoot database issues.
- End user - These days, all modern web or mobile applications store user data. How do you think they do it? Yes, applications are programmed in such a way that they collect user data and store the data on a DBMS system

running on their server. End users are the ones who store, retrieve, update and delete data.

- The users of the database can be classified into different groups.
    i. Naive Users
    ii. End Users
    iii. Sophisticated User
    iv. Application Users
    v. DBA - Database Administrator

Advantages of DBMS:

1.Reduction of redundancies:

Centralized control of data by the DBA avoids unnecessary duplication of data and effectively reduces the total amount of data storage required avoiding duplication in the elimination of the inconsistencies that tend to be present in redundant data files.

2.Sharing of data:

A database allows the sharing of data under its control by any number of application programs or users.

3.Data Integrity:

Data integrity means that the data contained in the database is both accurate and consistent. Therefore data values being entered for storage could be checked to ensure that they fall with in a specified range and are of the correct format.

4.Data Security:

The DBA who has the ultimate responsibility for the data in the dbms can ensure that proper access procedures are followed including proper authentication schemas for access to the DBS and additional check before permitting access to sensitive data.

5.Conflict resolution:

DBA resolve the conflict on requirements of various user and applications. The DBA chooses the best file structure and access method to get optional performance for the application.

6.Data Independence:

Data independence is usually considered from two points of views; physically data independence and logical data independence.

Physical data Independence allows changes in the physical storage devices or organization of the files to be made without requiring changes in the conceptual view or any of the external views and hence in the application programs using the data base.

Logical data independence indicates that the conceptual schema can be changed without affecting the existing external schema or any application program.

7.Easy Maintenance: It can be easily maintainable due to the centralized nature of the database system.

Disadvantages of DBMS

1. **Cost of Hardware and Software:** It requires a high speed of data processor and large memory size to run DBMS software.
2. **Size:** It occupies a large space of disks and large memory to run them efficiently.
3. **Complexity:** Database system creates additional complexity and requirements.
4. **Higher impact of failure:** Failure is highly impacted the database because in most of the organization, all the data stored in a single database and if the database is damaged due to

electric failure or database corruption then the data may be lost forever.

DBMS Languages

1. Data Definition Language (DDL)

- **DDL** stands for **D**ata **D**efinition **L**anguage. It is used to define database structure or pattern.
- It is used to create schema, tables, indexes, constraints, etc. in the database.
- Using the DDL statements, you can create the skeleton of the database.
- Data definition language is used to store the information of metadata like the number of tables and schemas, their names, indexes, columns in each table, constraints, etc.

Here are some tasks that come under DDL-

- **Create:** It is used to create objects in the database.
- **Alter:** It is used to alter the structure of the database.
- **Drop:** It is used to delete objects from the database.
- **Truncate:** It is used to remove all records from a table.

- **Rename:** It is used to rename an object.
- These commands are used to update the database schema that's why they come under Data definition language.

2. Data Manipulation Language (DML)

- It is used for accessing and manipulating data in a database. It handles user requests.
- Here are some tasks that come under DML:
- **Select:** It is used to retrieve data from a database.
- **Insert:** It is used to insert data into a table.
- **Update:** It is used to update existing data within a table.
- **Delete**: It is used to delete all records from a table.
- **Merge:** It performs UPSERT operation, i.e., insert or update operations.

3. Data Control Language (DCL)

- It is used to retrieve the stored or saved data.
- The DCL execution is transactional. It also has rollback parameters.
- Here are some tasks that come under DCL:

- **Grant:** It is used to give user access privileges to a database.
- **Revoke:** It is used to take back permissions from the user.
- There are the following operations which have the authorization of Revoke:
- INSERT, USAGE, EXECUTE, DELETE, UPDATE and SELECT.

4.Transaction Control Language (TCL)

TCL is used to run the changes made by the DML statement. TCL can be grouped into a logical transaction.

Here are some tasks that come under TCL:

Commit: It is used to save the transaction on the database.

Rollback: It is used to restore the database to original since the last Commit.

Roles in the Database Environment

A database environment is a collection of components that regulates the use of data, management, and a group of data.

- The database and the DBMS are corporate resources that must be managed like any other resource.
- **The Data Administrator (DA)** is responsible for defining data elements, data names and their relationship with the database. They are also known as **Data Analyst**.
- **Database Administrators (DBA)**

  A person who has central control over the system is called database administrator .
- A Database Administrator (DBA) is an IT professional who works on creating, maintaining, querying, and tuning the database of the organization.
- They are also responsible for maintaining data security and integrity.
- Installing and upgrading the DBMS Servers, Design and implementation, Performance Tuning
- Backup & Recovery

- Documentation, Security
- Granting of authorization for data access

  A good performing database is in the hands of DBA.

➢**Database Designers**

✔**Logical Database Designers**

- The logical database designer is concerned with identifying the data (that is, the entities and attributes), the relationships between the data, and the constraints on the data that is to be stored in the database.
- The logical database designer must have a thorough and complete understanding of the organization's data and any constraints on this data.

✔**Physical Database Designers**

- The physical database designer decides how the logical database design is to be physically realized.
- mapping the logical database design into a set of tables and integrity constraints.

- selecting specific storage structures and access methods for the data to achieve good performance.

Application Developers

- Once the database has been implemented, the application programs that provide the required functionality for the end-users must be implemented. This is the responsibility of the application developers.
- They are the developers who interact with the database by means of DML queries. These DML queries are written in the application programs like C, C++, JAVA, Pascal etc.

➢ **End Users**

- The end-users are the ‘clients’ for the database, which has been designed and implemented, and is being maintained to serve their information needs.

✔ **Sophisticated Users**

- The sophisticated end-user is familiar with the structure of the database and the facilities offered by the DBMS. The user passes each query to the query processor.

✔Naive Users

These are the users who use the existing application to interact with the database. For example, online library system, ticket booking systems, ATMs etc

Basic	DBMS Approach	File System Approach
Recovery Mechanism	DBMS provides a crash recovery mechanism, i.e., DBMS protects the user from system failure.	The file system doesn't have a crash mechanism, i.e., if the system crashes while entering some data, then the content of the file will be lost.
Manipulation Techniques	DBMS contains a wide variety of sophisticated techniques to store and retrieve the data.	The file system can't efficiently store and retrieve the data.
Concurrency Problems	DBMS takes care of Concurrent access of data using some form of locking.	In the File system, concurrent access has many problems like redirecting the file while deleting some information or updating some information.
Where to use	Database approach used to manage large amount of data.	File system approach used for small amount of data.

Basic	DBMS Approach	File System Approach
Cost	The database system is expensive to design.	The file system approach is cheaper to design.
Data Redundancy and Inconsistency	Due to the centralization of the database, the problems of data redundancy and inconsistency are controlled.	In this, the files and application programs are created by different programmers so that there exists a lot of duplication of data which may lead to inconsistency.
Structure	The database structure is complex to design.	The file system approach has a simple structure.
Data Independence	In this system, Data Independence exists, and it can be of two types. ➢Logical and Physical data Independence	In the File system approach, there exists no Data Independence.

Basic	DBMS Approach	File System Approach
Integrity Constraints	Integrity Constraints are easy to apply.	Integrity Constraints are difficult to implement in file system.
Data Models	In the database approach, 3 types of data models exist. ❖Hierarchal, Network and Relational data models	In the file system approach, there is no concept of data models exists.
Flexibility	Changes are often a necessity to the content of the data stored in any system, and these changes are more easily with a database approach.	The flexibility of the system is less as compared to the DBMS approach.
Examples	Oracle, SQL Server, Sybase etc.	Cobol, C++ etc.